

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	T. Kiran Kumay	Reg. No.:	192472321
Section / Batch:	CSE AI (2024)	Date:	

Code of Conduct

I T. Kiran Kumay / 192472321 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Kiran Kumay

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

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Section 1: Regular Expression Pattern Generation

For the given student details formulate Regex patterns for the following:

Email ID :

Pattern

$^[a-zA-Z0-9._\%+-]+\@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}\$$

Mobile Number :

Pattern

$^[6-9]\d{9}\$$

Strong Password

Pattern

$^(?=.*[A-Z])(?=.*[a-z])(?=.*[0-9])(?=.*[@\#\$\%\&!]).{8,}\$$

Date of Birth

Pattern

$^\d{2}\d{2}\d{4}\$$

Register Number

Pattern

$^\d{2}\d{2}\d{2}\d{3}\$$

Section 2: Search for a String:

Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand.

1. Write a Python program using the re module to perform the following operations for the word "AI":

Code:

```
import re
```

```
text = """Artificial Intelligence (AI) is transforming industries across the world.
```

```
AI is used in healthcare to assist doctors in diagnosis.
```

```
Many companies invest heavily in AI research because AI improves efficiency."""
```

```
result = re.search(r'AI', text)
```

```
if result:
```

```
    print("AI found")
```

```
else:
```

```
    print("AI not found")
```

Output:

AI found

2. Display the starting position of the first occurrence.

Code:

```
result = re.search(r'AI', text)
```

```
if result:
```

```
    print("Starting Position:", result.start())
```

Output:

Starting Position: 25

3. Display the ending position of the first occurrence.

```
result = re.search(r'AI', text)
```


if result:

```
print("Ending Position:", result.end())
```

Output:

Ending Position: 27

4. Display the total number of times the word "AI" appears in the passage

Code:

```
count = len(re.findall(r'AI', text))  
print("Total Occurrences:", count)
```

Output:

Total Occurrences: 4

5. Print a message if the word is not found

Code:

```
result = re.search(r'Python', text)
```

if result:

```
    print("Word found")
```

else:

```
    print("Word not found")
```

Output:

Word not found

Section 3: Split the documentation into sentences and words

1. Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.

import re

```
text = """Artificial Intelligence (AI) is transforming industries across the world. AI is used in  
healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to  
provide personalized learning experiences. Many companies invest heavily in AI research  
because AI improves efficiency and enables intelligent decision-making. As AI continues to  
evolve, professionals with AI skills are in high demand."""
```

```
sentences = re.split(r'[.!?]+', text)
```

```
print(sentences)
```

Output:

['Artificial Intelligence (AI) is transforming industries across the world',

'AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences',

'Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making',

'As AI continues to evolve, professionals with AI skills are in high demand',

"]

2.Count and display the total number of sentences.

Code:

```
print(len(sentences)-1)
```

Output:

4

3. Split the passage into words by considering one or more whitespace characters as delimiters.

Code:

```
words = re.split(r'\s+', text)
```

```
print(words)
```

Output:

```
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming',  
'industries', 'across', 'the', 'world.', 'AI', 'is', 'used',  
'in', 'healthcare', 'to', 'assist', 'doctors', 'in',  
'diagnosis', 'in', 'banking', 'to', 'detect', 'fraud',  
'and', 'in', 'education', 'to', 'provide', 'personalized',  
'learning', 'experiences.', 'Many', 'companies', 'invest',  
'heavily', 'in', 'AI', 'research', 'because', 'AI',  
'improves', 'efficiency', 'and', 'enables', 'intelligent',  
'decision-making.', 'As', 'AI', 'continues', 'to', 'evolve',  
'professionals', 'with', 'AI', 'skills', 'are', 'in',  
'high', 'demand.']
```

4. Count and display the total number of words.

Code:

```
print(len(words))
```

Output:

60

5. Display the list of extracted sentences and words.

```
print("Sentences:", sentences)
```

```
print("Words:", words)
```

Output:

Sentences:

1. Artificial Intelligence (AI) is transforming industries across the world
2. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences
3. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making
4. As AI continues to evolve, professionals with AI skills are in high demand

Words:

```
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', ..., 'demand.']
```

Section 4: Email and Strong Password Validation

A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Q1. Validate Email ID

Code:

```
import re
```

```
email = "john123@gmail.com"
```

```
pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.(com|org|edu|in)$'
```

```
if re.fullmatch(pattern, email):
```

```
    print("Valid Email")
```

```
else:
```

```
    print("Invalid Email")
```


Output:

Valid Email

Q2. Validate Strong Password

Code:

```
import re
```

```
password = "P@ssw0rd123"
```

```
pattern = r'^(?=.*[A-Z])(?=.*[a-z])(?=.*[0-9])(?=.*[@#$%&!])\S{8,}$'
```

```
if re.fullmatch(pattern, password):
```

```
    print("Strong Password")
```

```
else:
```

```
    print("Weak Password")
```

Output:

Strong Password

Section 5: Compile Once, Validate Many

A company receives 1,000 email IDs from its customers and needs to validate each one efficiently. Instead of compiling the same regular expression for every email, write a Python program using the `re.compile()` method to compile the pattern once and reuse it to validate all the email IDs.

Code:

```
import re
```

```
emails = [
```

```
    "john@gmail.com",
```

```
    "abc@yahoo.in",
```

```
    "student@college.edu",
```

```
    "test123@org.org",
```

```
    "wrongemail@"
```

```
]
```

```
pattern = re.compile(
```

```
    r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.(com|org|edu|in)$'
```

)

for email in emails:

```
if pattern.fullmatch(email):
```

```
    print(email, "-> Valid")
```

```
else:
```

```
    print(email, "-> Invalid")
```

Output:

john@gmail.com -> Valid

abc@yahoo.in -> Valid

student@college.edu -> Valid

test123@org.org -> Valid

wrongemail@ -> Invalid

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1						20	100
2	8	8	8	8	8	20	
3	8	8	8	8	8	20	
4	8	8	8	8	8	20	
5						20	

Faculty In-charge	Course Coordinator	Program Director
Dr. P. Kumaresan		
Signature: _____	Signature: _____	Signature: _____
Date: 24/12/20	Date: _____	Date: _____